

KARNATI MOHITH REDDY

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with hands-on experience building full-stack applications, REST APIs, AI/ML solutions, and LLM-powered systems. Experienced with Python, Django, FastAPI, React, SQL/NoSQL databases, and Gemini APIs, with practical work in RAG, vector search, authentication, and real-time applications. Built and tested end-to-end applications combining backend services, databases, and Generative AI.

EDUCATION

Anurag University, Hyderabad, Telangana

2022 – 2026

Bachelor of Technology in Computer Science and Engineering | CGPA: 7.29/10

PROJECTS

Multi-Tenant SaaS Platform | TenantVault **Live Demo:** <https://multi-tenant-saa-s-nine.vercel.app>

- Developed a multi-tenant SaaS platform with 3 subscription tiers, tenant-scoped JWT authentication, 15+ REST APIs, and 8 database models.
- Reduced document processing response time to under 200ms for files up to 100MB by offloading AES-128 encryption to async Celery/Redis workers with zero-knowledge architecture.
- Enabled real-time collaborative editing via Django Channels and WebSockets, delivering sub-second live cursor and keystroke sync across concurrent users.

GitHub: <https://github.com/mohithreddy123-hub/Multi-tenant-SaaS>

AI-Based Contactless Vital Sign Monitoring System

- Developed a non-contact vital sign monitoring system using rPPG and POS algorithm to extract HR and respiratory rate from facial RGB signals via standard webcam.
- Implemented FFT-based frequency estimation with bandpass filters (0.82.5 Hz for HR, 0.10.5 Hz for RR) and a Gradient Boosting Regressor trained on 9 signal features.
- Built a Streamlit app with real-time face detection, forehead ROI extraction, signal quality scoring, health suggestions, and PDF report generation.

GitHub: <https://github.com/mohithreddy123-hub/FaceVitalsPrototype>

QueryMind | Text-to-SQL RAG Assistant

- Built a 5-stage Text-to-SQL RAG pipeline using ChromaDB and Gemini embeddings to convert natural-language questions into executable MySQL queries through dynamic schema retrieval.
- Implemented two-tier SQL security combining application-layer validation with a read-only MySQL user, blocking DDL/DML commands, injection patterns, and comment-based attacks.
- Achieved 57/57 passing Pytest tests covering mocked LLM calls, live ChromaDB retrieval, SQL validation, and end-to-end pipeline execution.

GitHub: <https://github.com/mohithreddy123-hub/QueryMind>

TECHNICAL SKILLS

Languages: Python, JavaScript

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Streamlit

Backend: Django, Django REST Framework, FastAPI, Django Channels

Databases: PostgreSQL, MySQL, Redis

DevOps & Cloud: Git, GitHub, Vercel, Render, Cloudinary

Architecture: REST APIs, JWT Auth, RBAC, Multi-Tenancy, WebSockets, Async Programming

Data & ML: NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib

Generative AI: Google Gemini API, LLM API Integration, Prompt Engineering, Structured JSON Output, RAG, Vector Databases (ChromaDB), Embedding Models, LangChain

CERTIFICATIONS

Microsoft Azure AI Fundamentals: Azure AI tools and responsible AI practices

Oracle Certified Professional: OCI Generative AI Professional (2025)

AI Fluency: Framework & Foundations: Anthropic (2025)

Cisco Python Essentials 1 & 2: Cisco Networking Academy (May 2025)

AWS Cloud Practitioner Essentials: AWS Training & Certification (2026)